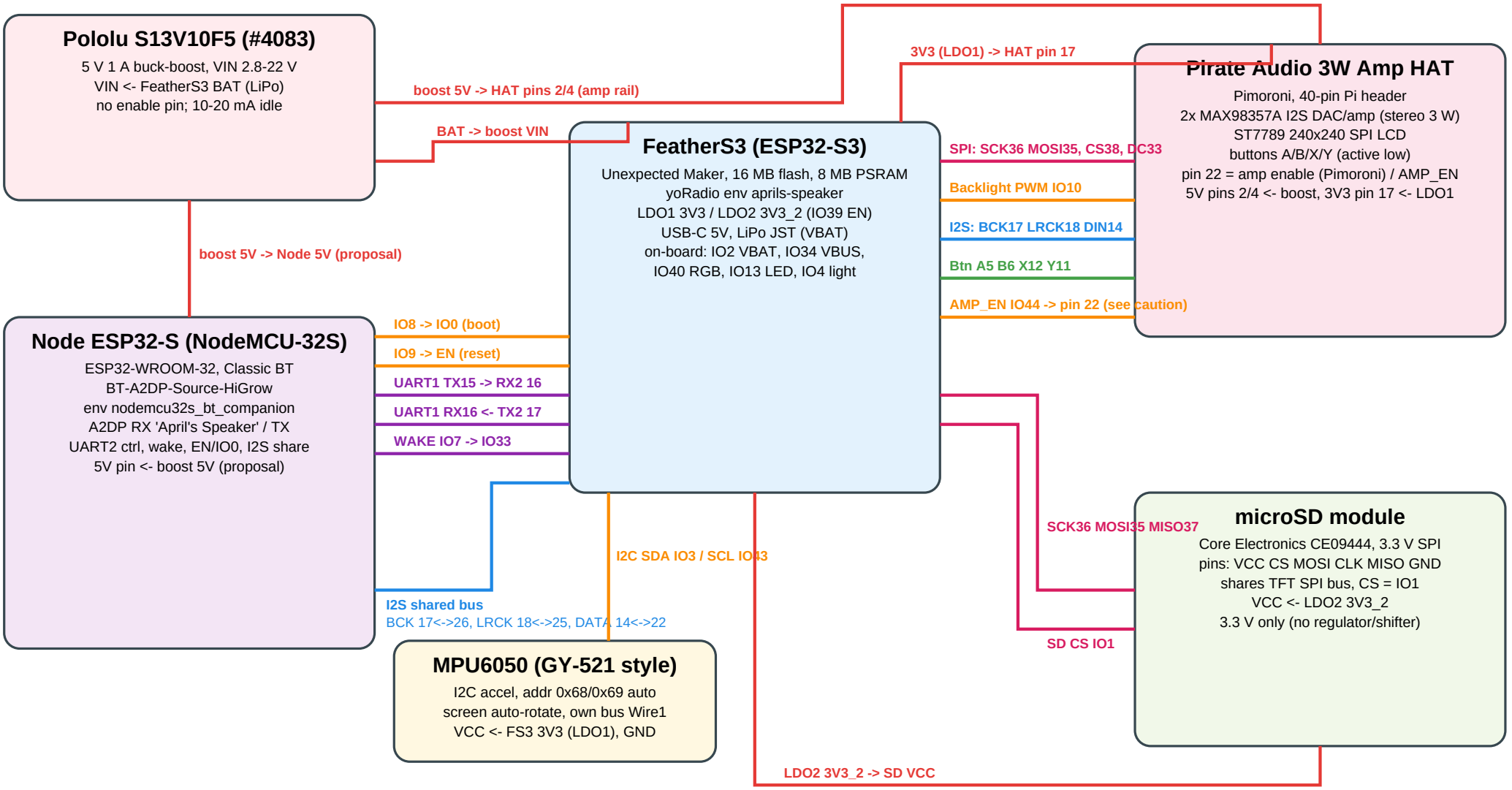
April's Speaker - overall block wiring diagram

SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | Caution / verify

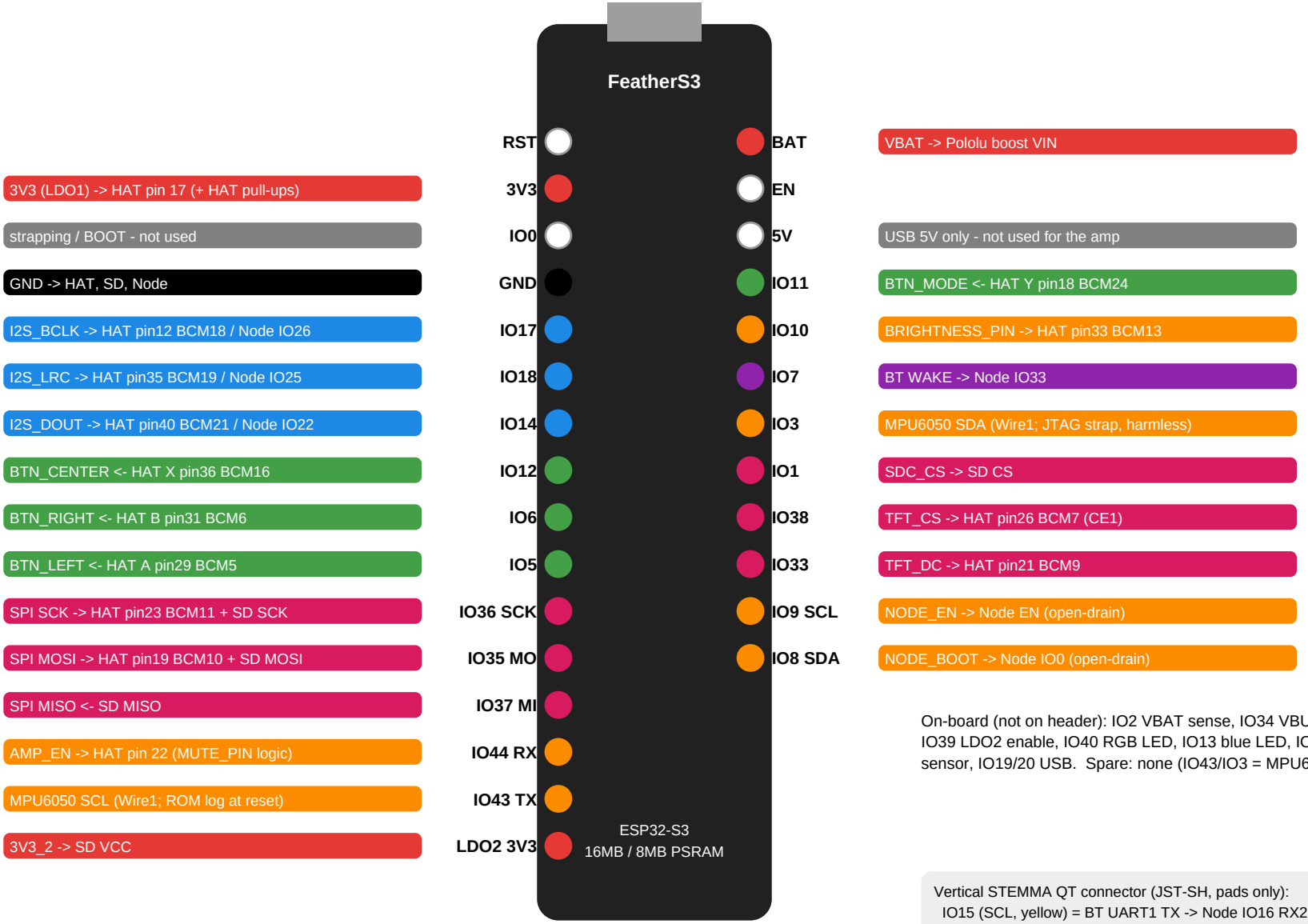


Ground: one common GND between FeatherS3, HAT (pins 9/25/39), SD module and Node ESP32.
All signal lines are 3.3 V logic on every part; no level shifting needed between them.
BT companion is ACTIVE (BT_COMPANION_PINS_TODO removed). Node EN/IO0 are open-drain from the S3: reset + flash the Node over the FeatherS3 USB.
CAUTION pin 22: Pimoroni documents it as a 3.3 V amp ENABLE input, not power. Verify before putting 5 V on it (see page 8).
I2S is shared: only one side may drive BCK/LRCK/DATA at a time (firmware handover, as on the ProS3).

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

April's Speaker - FeatherS3 pinout (top view, USB-C at top)

SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | Caution / verify



On-board (not on header): IO2 VBAT sense, IO34 VBUS sense, IO39 LDO2 enable, IO40 RGB LED, IO13 blue LED, IO4 light sensor, IO19/20 USB. Spare: none (IO43/IO3 = MPU6050).

Vertical STEMMA QT connector (JST-SH, pads only):
IO15 (SCL, yellow) = BT UART1 TX -> Node IO16 RX2
IO16 (SDA, blue) = BT UART1 RX <- Node IO17 TX2
red = LDO2 3V3: do NOT connect to the Node; black = GND

Physical order verified against Unexpected Maker 'FeatherS3_Pin_Reference.png' and kicad/symbols/FeatherS3.kicad_sym (github.com/UnexpectedMaker/esp32s3).
Firmware defines: boards/aprils_speaker.h. TFT_RST = -1 (HAT has no reset line).

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

April's Speaker - Pirate Audio 3W Amp HAT - 40-pin header (Pi numbering, pin 1 top-left)

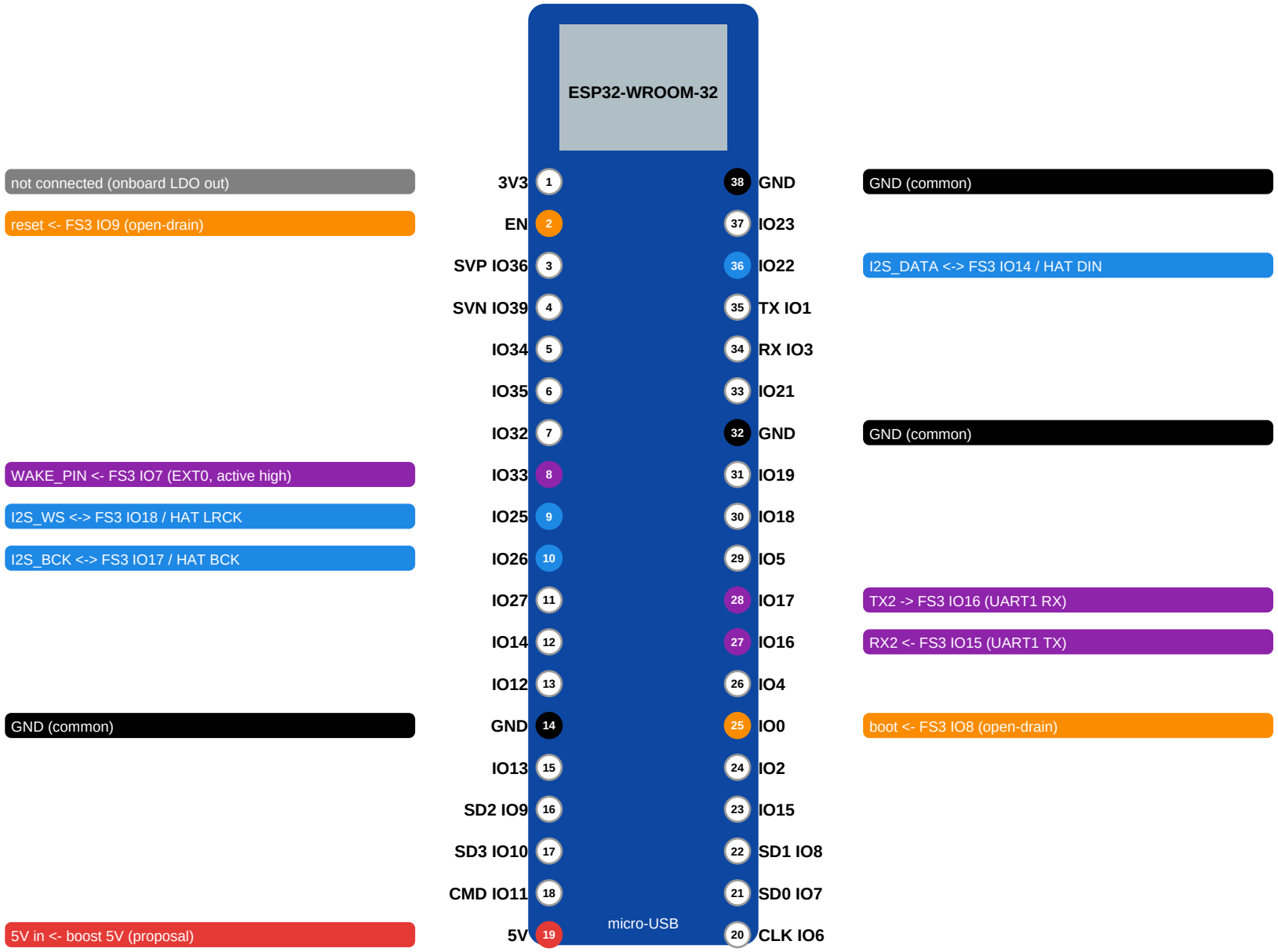
SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | Caution / verify

Pin roles verified against pinout.xyz 'pimoroni-pirate-audio-3w-stereo-amp' (power 2,17; ground 9,25,39; I2S 12/35/40; amp enable 22 active high; buttons 29/31/36/18 (old boards Y = 38); LCD 33 backlight, 21 D/C, 19 MOSI, 23 SCLK, 26 CS) and the Pimoroni product page (2x MAX98357A, ST7789 240x240, buttons active low BCM 5/6/16/24, 'DAC enable pin BCM 25 must be driven high'). Unused pins are white.
CAUTION pin 22: drawn as Pimoroni documents it (3.3 V logic enable, driven by FS3 IO44). Cain thinks it may be the amp power pin - measure/trace it before applying 5 V there: 5 V on a 3.3 V enable could damage the amp, and IO44 must never see 5 V (use a MOSFET/level stage if pin 22 really needs power).

Source: yoRadio-Fusion-FeS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

April's Speaker - Node ESP32-S (NodeMCU-32S, 38-pin) - top view, antenna at top

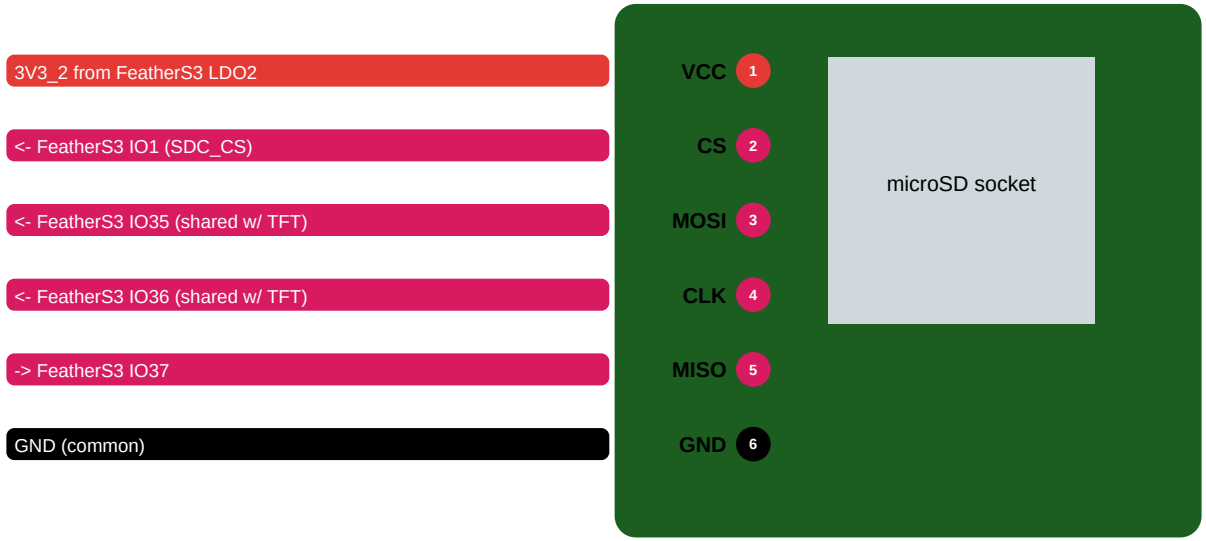
SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | Caution / verify



Physical order verified against the Ai-Thinker 'NodeMCU-32S specification v1.3' pin table (pins 1-19 left, 20-38 right from the USB end). Firmware: include/options.h COMPANION_BOARD_NODEMCU32S block. Avoided: flash pins 6-11, straps 0/2/5/12/15, UART0 1/3. Board confirmed: 38-pin (2x19) NodeMCU-32S.
I2S pins 25/26/22 are inputs (high-Z) except while the Node is the active I2S master (BT receive): set at boot and after every handover.

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

SPI (TFT/SD) | I2S audio | Buttons | BT companion | Power | Ground | Control | On-board fixed | Caution / verify



Pin order from Core Electronics (product CE09444): top to bottom with the silkscreen arrow pointing up: 1 VCC (3V3), 2 CS, 3 MOSI, 4 CLK, 5 MISO, 6 GND.
Power and logic are 3.3 V only (no regulator / level shifter): never feed it 5 V. Firmware: SDC_CS 1, SPI 36/35/37, SDSPISPEED 10 MHz, SD_SHARE_TFT_SPI 1.
Recommended: ~100 R series resistors on the SD lines and a 10 k pull-up on CS (ProS3 practice).
Firmware drives SDC_CS and TFT_CS HIGH at boot before SPI starts (SD park) and powers LDO2 (IO39) first.

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

April's Speaker - master connection table

#	From (part / pin)	To (part / pin)	Signal	Notes
1	FeatherS3 IO36 (SCK)	HAT pin 23 (BCM11) + SD CLK (4)	SPI SCLK	shared TFT/SD bus, TFT 20 MHz / SD 10 MHz; 3.3 V
2	FeatherS3 IO35 (MO)	HAT pin 19 (BCM10) + SD MOSI (3)	SPI MOSI	shared
3	SD MISO (5)	FeatherS3 IO37 (MI)	SPI MISO	HAT does not use MISO
4	FeatherS3 IO38	HAT pin 26 (BCM7, CE1)	TFT_CS	driven HIGH at boot
5	FeatherS3 IO33	HAT pin 21 (BCM9)	TFT_DC	
6	-	HAT (LCD reset)	TFT_RST	none: TFT_RST -1, HAT has no reset line
7	FeatherS3 IO10	HAT pin 33 (BCM13)	Backlight PWM	BRIGHTNESS_PIN (LEDC)
8	FeatherS3 IO1	SD CS (2)	SDC_CS	driven HIGH at boot (SD park); 10 k pull-up recommended
9	FeatherS3 IO17	HAT pin 12 (BCM18) + Node IO26	I2S BCLK	shared with companion; only one side drives, the other is high-Z
10	FeatherS3 IO18	HAT pin 35 (BCM19) + Node IO25	I2S LRCK/WS	
11	FeatherS3 IO14	HAT pin 40 (BCM21) + Node IO22	I2S DATA	no MCLK (MAX98357A needs none)
12	FeatherS3 IO44	HAT pin 22 (BCM25)	AMP_EN	AMP_EN_PIN = MUTE_PIN (MUTE_VAL LOW): low when stopped/asleep, high when playing + BT receive. 3.3 V logic: VERIFY pin 22 before any 5 V; MOSFET optional
13	HAT pin 29 (BCM5) Btn A	FeatherS3 IO5	BTN_LEFT	active low; HAT pull-ups + internal pull-up; RTC GPIO
14	HAT pin 31 (BCM6) Btn B	FeatherS3 IO6	BTN_RIGHT	active low

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

#	From (part / pin)	To (part / pin)	Signal	Notes
15	HAT pin 36 (BCM16) Btn X	FeatherS3 IO12	BTN_CENTER	active low
16	HAT pin 18 (BCM24) Btn Y	FeatherS3 IO11	BTN_MODE	unchanged from current firmware; deep-sleep wake (WAKE_PIN1)
17	FeatherS3 IO15 (STEMMA QT 2 SCL)	Node IO16 (RX2)	BT UART1 TX	115200 8N1; off UART0 so the S3 ROM boot log never reaches the Node
18	Node IO17 (TX2)	FeatherS3 IO16 (STEMMA QT 2 SDA)	BT UART1 RX	115200; also the Node-flash bridge (baud follows esptool)
19	FeatherS3 IO7	Node IO33	BT WAKE	EXT0 active high, 600 ms pulse, pulled down on Node
20	FeatherS3 IO9	Node EN (pin 2)	NODE_EN	open-drain: driven low to reset, released otherwise (Node 10 k pull-up)
21	FeatherS3 IO8	Node IO0 (pin 25)	NODE_BOOT	open-drain: low only while entering the Node ROM loader
22	FeatherS3 3V3 (LDO1)	HAT pin 17	3V3	LCD logic + button pull-ups; LDO1 stays on in deep sleep
23	FeatherS3 LDO2 3V3_2	SD VCC (1)	3V3_2	off in deep sleep; LDO2 EN = IO39. Module is 3.3 V only
24	FeatherS3 BAT	Pololu S13V10F5 VIN	VBAT	2.8-22 V in; no enable pin, 10-20 mA idle drain (add a switch if it sits on battery)
25	Pololu VOUT (5 V)	HAT pins 2 + 4	5V amp rail	1 A limit (S13V15F5 / S13V20F5 are pin-compatible upgrades)
26	Pololu VOUT (5 V)	Node 5V (pin 19)	Node power	PROPOSAL (not confirmed); shares the 1 A budget with the amp
27	FeatherS3 GND	HAT 6/9/14/20/25/30/34/39; SD GND (6); Node GND; Pololu GND; MPU GND	GND	single common ground
28	FeatherS3 IO3	MPU6050 SDA	ORIENT_SDA	Wire1 (own bus), 100 kHz; module pull-ups OK (IO3 strap only read if eFuse selects JTAG)
29	FeatherS3 IO43	MPU6050 SCL	ORIENT_SCL	Wire1; ROM prints its boot banner on IO43 at reset (harmless glitch, I2C takes the pin at start)
30	FeatherS3 3V3 (LDO1)	MPU6050 VCC	3V3	3.3 V; MPU INT and AD0 not connected (address auto-detected 0x68/0x69)
-	FeatherS3 on-board	IO2 VBAT, IO34 VBUS, IO39 LDO2 EN, IO40 RGB, IO13 LED, IO4 light	fixed	not wired externally

Source: yoRadio-Fusion-ProS3 branch cursor/aprils-speaker-pins (boards/aprils_speaker.h, platformio.ini env aprils-speaker)
+ BT-A2DP-Source-HiGrow branch cursor/aprils-speaker-oct23 (include/options.h, env nodemcu32s_bt_companion). Pins assigned 24 Oct 2026 (in code); not yet bench-tested.

POWER & GROUND
- FeatherS3: USB-C 5 V or 1S LiPo (JST). LDO1 3V3 always on; LDO2 3V3_2 enabled by IO39, off in deep sleep (also powers the RGB LED and the vertical STEMMA QT).
- Pololu S13V10F5 (#4083): 5 V 1 A buck-boost, VIN from FeatherS3 BAT, VOUT to HAT 5V pins 2/4 (amp rail) and, as a proposal, the Node 5V pin. No enable pin.
- HAT 3V3 (pin 17) from LDO1 so button pull-ups stay valid in deep sleep (Y wakes). SD adapter on LDO2. One common ground. All logic is 3.3 V.

CAUTION - HAT pin 22
Pimoroni / pinout.xyz document pin 22 (BCM25) as the amp ENABLE, 3.3 V logic, active high - not a power input. Cain's reading is that it is the amp power pin.
Drawn here as an enable driven by FeatherS3 IO44 (AMP_EN). Measure or trace it before putting 5 V on it. If it really needs power, switch the boost 5 V with a
MOSFET driven by IO44 - never connect IO44 directly to 5 V.

UPDATING THE NODE THROUGH THE FEATHERS3 USB (cover stays on)
1. curl 'http://<speaker-ip>/set?nodeflash=1' (or type 'nodeflash 1' on the USB serial console). The S3 reboots into a bridge: no WiFi/audio, screen stays dark.
2. It holds Node IO0 low, pulses EN (Node enters its ROM loader) and bridges USB <-> UART1. Then, with the FeatherS3 port:
   esptool --chip esp32 --port /dev/cu.usbmodemXXXX --before no-reset --after no-reset -b 460800 write-flash 0x10000 .pio/build/nodemcu32s_bt_companion/firmware.bin
   (--before no-reset is REQUIRED: the default reset would reset the FeatherS3 itself. Baud changes are followed automatically.)
3. 30 s after the last traffic (5 min if esptool never connects) the S3 resets the Node into its app and restarts normally. Power-cycling also exits.

SOURCES
- Unexpected Maker FeatherS3: https://esp32s3.com/feathers3.html ; https://github.com/UnexpectedMaker/esp32s3 (pin card, KiCad symbol)
- Pimoroni Pirate Audio 3W Stereo Amp: https://shop.pimoroni.com/products/pirate-audio-3w-stereo-amp ; https://pinout.xyz/pinout/pirate_audio_3w_amp
- Ai-Thinker NodeMCU-32S specification v1.3: https://component.aithinker.com/_media/nodemcu32-s_specification_v1.3.pdf
- Core Electronics microSD Adapter CE09444: https://core-electronics.com.au/microsd-adapter.html
- Pololu S13V10F5 #4083: https://www.pololu.com/product/4083
- Firmware: yoRadio-Fusion-ProS3 boards/aprils_speaker.h, src/core/node_flash.cpp, docs/APRILS_SPEAKER.md; BT-A2DP-Source-HiGrow include/options.h